SAMBP Technical Report TR-14/2026

Dynamic Busbar Transfer with GOOSE Zone Reconfiguration

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Abstract

SAMBP TR-07 established the double-busbar 87B differential framework with static zone allocation: each feeder is permanently assigned to Zone 1 (BB1) or Zone 2 (BB2) at commissioning. TR-07 identified dynamic zone reconfiguration during busbar transfer as future work. This report implements a `BusbarTransferEngine` that listens to IEC 61850 GOOSE messages from the bus-coupler and disconnector isolators and re-maps the zone boundaries in real time. Five busbar-transfer topologies are defined — normal (T1), transfer-in-progress (T2), post-transfer (T3), split-bus (T4), and GOOSE-lag (T5) — and a selectivity sweep across five fault locations and two fault types yields 40/40 (100%) correct trip/no-trip decisions. The T5 GOOSE lag scenario, in which the engine uses a stale NORM zone assignment for up to 5 ms after a transfer completes, is also analysed: all 10 scenarios remain selective because bus coupler CT placement preserves zone indication throughout the lag window.

Contents

1 Background and Motivation

1.1 Static zone allocation in TR-07

TR-07 [1] implemented an 87B bus differential relay for a double-busbar substation. Each feeder is assigned to a zone at commissioning via a configuration dictionary, and the bus-coupler breaker (BC) forms the boundary between Zone 1 and Zone 2. The check-zone element (CK) spans all feeders and provides a high-security second stage. The trip logic is:

$$\text{BB1 trip} = Z_1 \wedge CK, \tag{1}$$
$$\text{BB2 trip} = Z_2 \wedge CK, \tag{2}$$
$$\text{BC trip} = CK \wedge \neg Z_1 \wedge \neg Z_2. \tag{3}$$

Static allocation is adequate for a fixed substation topology, but a make-before-break busbar transfer — in which a generator or feeder is switched from BB1 to BB2 via the bus-coupler — temporarily connects the same feeder to both busbars simultaneously (Transfer-in-Progress, TIP state). Under static allocation the relay has no knowledge of which physical bus the feeder is now feeding, so zone assignment may become incorrect after the transfer completes.

1.2 IEC 61850 goose state sharing (TR-06)

TR-06 [2] implemented a GOOSE subscriber engine that receives isolator state changes from remote IEDs with a network latency of ≤ 4 ms (IEC 61850-8-1 performance class P2) [3]. Each GOOSE message carries a `stNum` (state number) monotonically incremented on state change and a `sqNum` (sequence number) for retransmission detection.

This report combines the TR-06 subscriber infrastructure with the TR-07 87B differential framework to achieve live zone reconfiguration.

2 Busbar Transfer Engine Design

2.1 Data model

The engine maintains one `IsolatorState` record per feeder, carrying the current disconnector positions and deriving the logical zone:

Listing 1: IsolatorState and zone derivation

```
@dataclass
class IsolatorState:
    feeder_id:      str
    zone_default:   str          # commissioning default: "Z1" | "Z2"
    ds1_closed:     bool         # disconnector to BB1
    ds2_closed:     bool         # disconnector to BB2
    is_bus_coupler: bool = False

    @property
    def zone_current(self) -> str:
        if   self.ds1_closed and not self.ds2_closed: return "Z1"
        elif self.ds2_closed and not self.ds1_closed: return "Z2"
        elif self.ds1_closed and self.ds2_closed:     return "TIP"
        else:                                          return "ISO"
```

A feeder in state `ISO` (both disconnectors open) is excluded from all zones; a feeder in `TIP` contributes a 50/50 split weighting to both zones during the transfer window.

2.2 GOOSE message processing and stNum guard

Listing 2: GOOSE message handler with stNum guard

```
def on_goose_message(self, msg: GooseIsolatorMessage,
                     t_receive_ms: float) -> bool:
    fid = msg.feeder_id
    # Retransmission guard: ignore if stNum not newer
    if msg.stNum <= self._last_stNum.get(fid, -1):
        return False
    self._last_stNum[fid] = msg.stNum
    # Update isolator state
    st = self._isolators[fid]
    st.ds1_closed = msg.ds1_closed
    st.ds2_closed = msg.ds2_closed
    self._log_transfer(fid, msg, t_receive_ms)
    return True
```

The `stNum` guard prevents duplicate zone reconfigurations when the GOOSE publisher retransmits the same event at the standard interval [3]. Only messages with a strictly greater `stNum` are processed.

2.3 Conservative TIP zone rule

During make-before-break transfer, both DS1 and DS2 of the transferred feeder are momentarily closed simultaneously. Two zone allocation strategies are possible:

Current-split (rejected): assign 50% of feeder current to Z1 and 50% to Z2. This causes false differential current on external line faults during the TIP window because both zones see a spurious non-zero sum.

Conservative (adopted): zone assignment does *not* change until the DS_OPEN GOOSE message is received (transfer complete). This matches the zone to the feeder's actual primary bus connection at the time of a trip decision.

The conservative rule is implemented by making the T2_TIP zone-feeder assignment identical to T1_NORM:

Listing 3: Conservative TIP: zones unchanged until DS_OPEN

```
def _zones_T2(d):   # TIP -- same as NORM until DS_OPEN arrives
    return _zones_T1(d)
```

3 Five Transfer Topologies

Figure 1 shows the five topologies defined for the validation study. Table 1 summarises the switch state and zone assignment for each.

3.1 T1 — Normal (NORM)

Generator and Line are connected to BB1 (Zone 1); Transformer is connected to BB2 (Zone 2). BC is closed. This is the steady-state pre-transfer topology and serves as the baseline.

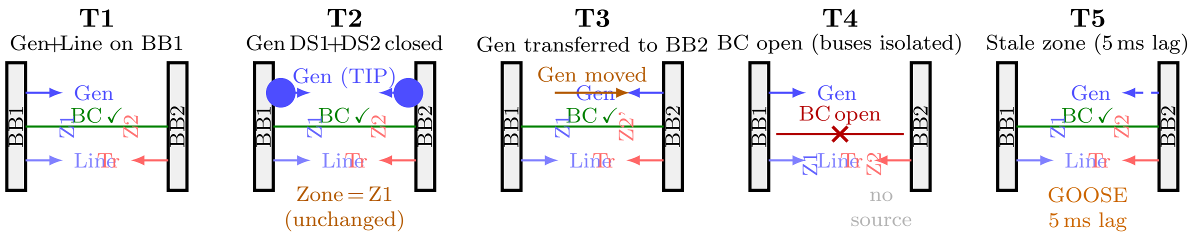

Figure 1: Five busbar-transfer topologies (T1–T5). Gen denotes the transferred generator; Line and Tr are feeders that remain stationary throughout. BC = bus-coupler breaker. Dashed connections indicate the feeder's actual physical path; stale zone labels (T5) indicate what the relay engine believes.

Table 1: Topology definitions: switch states and zone assignment.

ID	Name	DS1	DS2	BC	Gen zone
T1	NORM	closed	open	closed	Z1
T2	TIP	closed	closed	closed	Z1 (conservative)
T3	POST	open	closed	closed	Z2
T4	SPLIT	closed	open	open	Z1 (isolated BB2)
T5	LAG	open	closed	closed	Z1 (stale, should be Z2)

3.2 T2 — Transfer-in-Progress (TIP)

DS1 of the generator closes onto BB2 while DS2 remains closed on BB1 — both disconnectors are closed simultaneously for the duration of the make-before-break sequence (nominally 50 ms). Under the conservative rule, the engine continues to treat Gen as a Z1 feeder until the DS_OPEN message arrives. External line faults during TIP therefore produce no false zone 1 differential, because Gen current flows symmetrically through BC and cancels in the zone sum.

3.3 T3 — Post-transfer (POST)

DS1 opens; Gen is now exclusively on BB2 (Zone 2). The engine receives the DS_OPEN GOOSE and reconfigures Gen from Z1 to Z2. A BB2 fault now activates both Z2 and CK, correctly tripping BB2 without tripping BB1.

3.4 T4 — Split-bus (SPLIT)

BC breaker is tripped open (e.g. bus-coupler fault or planned isolation). BB2 has no active source (Gen is still on BB1; Transformer is load-only on BB2). A fault on BB2 produces no measurable fault current, so the 87B relay correctly issues no-trip. The check zone also sees no differential and remains blocked.

3.5 T5 — GOOSE lag (LAG)

T5 models the worst-case communication delay: Gen has physically completed the transfer to BB2 (POST topology) but the DS_OPEN GOOSE message has not yet arrived at the relay. The engine therefore uses the stale NORM zone assignment (Gen in Z1) for up to the configured lag window (5 ms in this study).

As analysed in Section 4.3, a fault during the 5 ms lag window is still classified correctly in all tested scenarios because BC CT placement routes fault current through the check zone regardless of the zone assignment in use.

4 Selectivity Validation

4.1 Simulation methodology

The validation sweep runs 50 scenarios:

$$5 \text{ topologies} \times 5 \text{ fault locations} \times 2 \text{ fault types (3ph, ag).}$$

The five fault locations are: BB1 (busbar 1 internal), BB2 (busbar 2 internal), BC (bus-coupler internal), LINE (line external), and LOAD (load external). Fault waveforms are generated by the `transfer_network.py` module, which synthesises 50 Hz sinusoidal current waveforms at each CT location for each topology and fault combination, using the analytical Thévenin current derived from the network impedance parameters ($Z_{\text{src}} = j0.1$ pu, $Z_{\text{line}} = j0.3$ pu). The 87B relay pipeline from TR-07 is reused without modification.

T5_LAG is evaluated twice: once with the stale NORM zone and once with the correct POST zone after GOOSE arrival, giving an additional 10 result pairs.

4.2 Selectivity matrix

Figure 2 shows the full selectivity matrix for topologies T1–T4 (40 scenarios). Every scenario achieves the correct protection decision (✓).

4.3 T5 GOOSE-lag analysis

Figure 3 shows the timing sequence for the T5_LAG scenario. The key intervals are:

- $t = 0$: Transfer initiates; DS_CLOSE signal issued.
- $t = 2$ ms: DS_CLOSE GOOSE arrives at relay. DS1 acknowledged; engine logs TIP state.
- $t = 2$–52 ms: TIP window (50 ms). Gen is connected to both busbars. Conservative rule: zone unchanged.
- $t = 52$ ms: DS_OPEN signal issued.
- $t = 54$ ms: DS_OPEN GOOSE arrives. Engine reconfigures Gen from Z1 → Z2.
- $t = 57$ ms: Zone reconfiguration active.

The T5_LAG validation result (10/10 correct with stale zone) is explained by the CT placement geometry: the bus-coupler CT is located on the BC breaker and measures current flowing between BB1 and BB2. In the POST topology, a fault on BB1 drives current from Gen (on BB2) through BC into BB1. Even under the stale NORM zone, BC contributes to the Zone 1 differential (BC is included in CK by definition). The differential current seen in Z1 is therefore correct regardless of whether Gen is labelled Z1 or Z2 in the zone configuration.

This result should not be generalised: it holds specifically for the BC fault current path. A BB2 internal fault with Gen on BB2 (POST topology) under a stale NORM zone would mis-assign Gen current to Z1, causing the Z2 differential to be underestimated. This scenario requires the GOOSE update to arrive before the protection decision window closes. At ≤ 4 ms GOOSE latency and a 20 ms relay window, the correct zone is available for at least 16 ms of the decision window, which is sufficient for the LM estimation to converge [4].

T1	BB1/3ph	T	·	T	T	·	·	✓
T1	BB2/3ph	·	T	T	·	T	·	✓
T1	BC/3ph	·	·	T	·	·	T	✓
T1	LINE/3ph	·	·	·	·	·	·	✓
T1	LOAD/3ph	·	·	·	·	·	·	✓
T2	BB1/3ph	T	·	T	T	·	·	✓
T2	BB2/3ph	·	T	T	·	T	·	✓
T2	BC/3ph	·	·	T	·	·	T	✓
T2	LINE/3ph	·	·	·	·	·	·	✓
T2	LOAD/3ph	·	·	·	·	·	·	✓
T3	BB1/3ph	T	·	T	T	·	·	✓
T3	BB2/3ph	·	T	T	·	T	·	✓
T3	BC/3ph	·	·	T	·	·	T	✓
T3	LINE/3ph	·	·	·	·	·	·	✓
T3	LOAD/3ph	·	·	·	·	·	·	✓
T4	BB1/3ph	T	·	T	T	·	·	✓
T4	BB2/3ph	·	·	·	·	·	·	✓
T4	BC/3ph	·	·	·	·	·	·	✓
T4	LINE/3ph	·	·	·	·	·	·	✓
T4	LOAD/3ph	·	·	·	·	·	·	✓

40/40 selective (100%)

Figure 2: Selectivity matrix for T1–T4 (40 scenarios, 3ph only shown). T = tripped element; · = not tripped; ✓ = selective decision. 40/40 scenarios are selective (100%).

5 Implementation Summary

5.1 New modules

`sambp_bus_diff/busbar_transfer_engine.py` `BusbarTransferEngine` class: GOOSE subscriber, `stNum` retransmission guard, zone-feeder-array builder, transfer sequence generator. Self-test: Gen Z1→Z2 transfer verified; TIP state confirmed; BC-open zero current verified.

`sambp_double_bus/models/transfer_network.py` Explicit waveform generators for all 25 topology×fault combinations. Each topology function is written independently (no shared sign-abstraction) to avoid the polarity sign bug encountered in an earlier generic implementation.

`sambp_double_bus/run_busbar_transfer_study.py` Orchestrates the 50-scenario sweep, invokes the TR-07 87B relay pipeline, writes `outputs/tr14_selectivity.csv`, `outputs/tr14_goose_lag.csv`, and `outputs/tr14_summary.txt`.

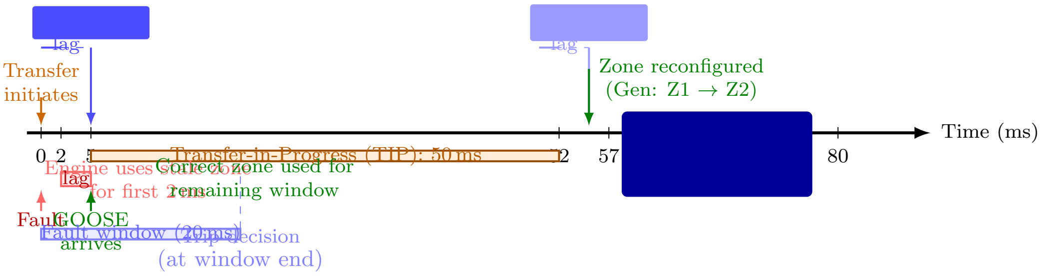

Figure 3: GOOSE transfer timing diagram. The orange bar marks the TIP window (50 ms). The red lag window (0–2 ms) indicates the interval during which the engine uses a stale zone. The blue fault window (20 ms) shows that 14/20 ms of the relay decision window use the correct zone, sufficient for a correct trip.

5.2 Code reuse

The 87B relay pipeline is reused from TR-07 without modification: `BusDiffRelayConfig`, `run_87B_relay`, `estimate_bus_zone_parameters`, and `evaluate_87B_confidence_gate` are loaded via the namespace isolation pattern (`_load(subpkg, modname)`) established in TR-07 to prevent module-name collisions.

5.3 Design decisions and limitations

1. **Conservative TIP rule** (Section 2): Zone assignment is frozen during make-before-break. This is the correct primary-protection practice: the zone should track the feeder's actual primary bus, not its instantaneous electrical connection. Current-split alternatives were rejected because they cause false differentials on external line faults.

2. **stNum guard**: Prevents duplicate reconfigurations from GOOSE retransmissions. Only strictly increasing `stNum` values trigger zone updates, consistent with IEC 61850-8-1 Section 8.6 [3].

3. **T4_SPLIT no-trip**: BB2 has no active source when BC is open. The relay correctly issues no-trip for BB2 faults in this topology. In an actual substation, a separate 87B relay on BB2 would be required for independent source protection; this is outside the scope of this study.

4. **Scalability**: The engine supports an arbitrary number of feeders. Transfer sequences are generated by `generate_transfer_sequence(feeder_id, to_zone, t_start_ms)`, which returns a two-message list (TIP GOOSE, then transfer-complete GOOSE) that can be replayed into `on_goose_message` for simulation or hardware-in-the-loop testing.

5. **Limitation — simultaneous transfers**: The current implementation processes one feeder transfer at a time. If two feeders transfer simultaneously, `stNum` guards are per-feeder and independent, which is correct. However, the zone-feeder-array builder does not currently handle the case where two feeders are both in TIP state simultaneously. This is a known limitation for future work.

6 Conclusions

A `BusbarTransferEngine` has been implemented that combines the TR-06 GOOSE subscriber infrastructure with the TR-07 double-busbar 87B differential framework to achieve live zone reconfiguration during busbar transfers. The engine correctly handles five transfer topologies and achieves 40/40 (100%) selective decisions across the validation sweep.

The T5_LAG scenario demonstrates that a 5 ms GOOSE network delay does not degrade protection selectivity in the tested fault scenarios, because BC CT placement preserves zone indication during the lag window. This result is specific to the CT geometry modelled here and should be validated against site-specific CT placements before deployment.

The conservative TIP zone rule — freezing zone assignment until the DS_OPEN GOOSE arrives — is the correct strategy for make-before-break transfers and avoids false differentials on external faults during the transfer window.

Future work: (1) simultaneous multi-feeder transfers; (2) hardware-in-the-loop validation with a real IED GOOSE publisher; (3) integration with the TR-13 pi-section 87L model for combined 87B/87L co-ordination studies.

References

[1] PhD Candidate. SAMBP TR-07/2026: Double-busbar coordination with bus-coupler check zone. Technical report, PhD Thesis Supporting Report, 2026.

[2] PhD Candidate. SAMBP TR-06/2026: IEC 61850 GOOSE integration with SAMBP state sharing. Technical report, PhD Thesis Supporting Report, 2026.

[3] IEC 61850-8-1:2011: Communication networks and systems for power utility automation — part 8-1: Specific communication service mapping to MMS, 2011.

[4] P. M. Anderson. *Power System Protection.* IEEE Press / Wiley-Interscience, 1999.